

Abstract algebra – 3

Permutations

13 сентября 2026 года

Theoretical minimum. Permutations are the most frequent algebraic subject at the Putnam and the IMC, and the only one that is used inside other topics: determinants, counting and probability all speak the language of S_n .

Definitions. S_n is the group of bijections of $\{1, \dots, n\}$ under composition, $|S_n| = n!$. A *cycle* $(i_1 i_2 \dots i_k)$ sends $i_1 \mapsto i_2 \mapsto \dots \mapsto i_k \mapsto i_1$ and fixes everything else; a *transposition* is a cycle of length 2. The *cycle type* of σ is the partition of n formed by the lengths of its cycles, fixed points counted as cycles of length 1.

Theorem 1 (cycle decomposition and conjugacy). Every permutation is a product of disjoint cycles, uniquely up to order: the cycles are the orbits of the cyclic group $\langle \sigma \rangle$ on $\{1, \dots, n\}$. Disjoint cycles commute, hence $\text{ord}(\sigma)$ is the least common multiple of the cycle lengths. Conjugation relabels: $\tau(i_1 \dots i_k) \tau^{-1} = (\tau(i_1) \dots \tau(i_k))$, so two permutations are conjugate in S_n exactly when they have the same cycle type, and the conjugacy classes correspond to partitions of n . The number of permutations with m_k cycles of length k equals $n! / \prod_k k^{m_k} m_k!$.

Theorem 2 (the sign). For $\sigma \in S_n$ let $\text{inv}(\sigma)$ be the number of pairs $i < j$ with $\sigma(i) > \sigma(j)$ and $c(\sigma)$ the number of cycles. Then

$$\text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)} = (-1)^{n-c(\sigma)},$$

and sgn is a homomorphism $S_n \rightarrow \{\pm 1\}$. *Proof:* multiplying by a transposition (ab) either splits one cycle into two or merges two cycles into one, so c changes by exactly 1 and the second expression changes sign; it also changes the parity of the number of inversions. Since transpositions generate S_n , the two expressions agree and both are multiplicative. The kernel is the alternating group A_n of index 2.

Theorem 3 (generation). Transpositions generate S_n ; the adjacent transpositions $(k, k+1)$ already suffice; and $(1\ 2)$ together with the long cycle $(1\ 2\ \dots\ n)$ generate S_n , because conjugating $(1\ 2)$ by the cycle produces all adjacent transpositions. The 3-cycles generate A_n , and A_n is simple for $n \geq 5$.

Fact 1 (permutations inside linear algebra). $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_i a_{i\sigma(i)}$, and the permanent is the same sum without signs. The permutation matrices P_σ form a subgroup of GL_n isomorphic to S_n , with $\det P_\sigma = \operatorname{sgn}(\sigma)$ and $\operatorname{tr} P_\sigma = \#\{\text{fixed points}\}$; this identity is the bridge between this sheet and Abstract algebra – 5.

Fact 2 (a random permutation). Let σ be uniform in S_n . The indicator of “ k is fixed” has mean $1/n$, so the expected number of fixed points is 1 for every n ; the distribution of that number converges to the Poisson law with parameter 1, and the probability of having no fixed point tends to $1/e$ by inclusion–exclusion. The expected number of cycles is $H_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} \sim \ln n$, which follows from the “ k -th element closes a cycle with probability $1/k$ ” construction.

1. Prove the uniqueness of the cycle decomposition, the formula $\operatorname{ord}(\sigma) = \operatorname{lcm}$ of the cycle lengths, and that conjugacy classes of S_n are exactly the cycle types. Then find the largest order of an element of S_{10} , and describe how the answer behaves for large n .
2. Prove that the parity of $\operatorname{inv}(\sigma)$ equals the parity of $n - c(\sigma)$, deduce that sgn is a homomorphism, and compute the sign of the permutation $k \mapsto n+1-k$ of $\{1, \dots, n\}$.
3. Let p be an odd prime with $p \equiv 2 \pmod{3}$, and define a permutation π of the residue classes modulo p by $\pi(x) \equiv x^3$. Show that π is an even permutation if and only if $p \equiv 3 \pmod{4}$. (Putnam 2012, B6)
4. For $\pi \in S_n$ let $\sigma(\pi) = \pm 1$ be its sign and $\nu(\pi)$ the number of its fixed

points. Show that

$$\sum_{\pi \in S_n} \frac{\sigma(\pi)}{\nu(\pi) + 1} = (-1)^{n+1} \frac{n}{n+1}.$$

(Putnam 2005, B6)

5. For a uniformly random $\sigma \in S_n$ compute the expected number of fixed points and the expected number of cycles, and prove that the probability of having no fixed point tends to $1/e$. What is the average number of local maxima of a permutation of $\{1, \dots, n\}$, where k is a local maximum if $\pi(k)$ exceeds its existing neighbours? (Putnam 2006, A4)